Circular Reasoning and Unsupported Statements

How to work:

- A reason must come from the **givens**, a **definition**, a **postulate**, or a theorem already proved. Anything else is not yet available to you.
- A step is **circular** when its reason is the very statement being proved (or a theorem whose proof depends on it).
- A step is **unsupported** when its reason describes a property nobody gave you — "the lines are parallel", "M is a midpoint", "those are vertical angles".
- Name the faulty step first, then repair it. Show the algebra for any measure you are asked to find.

1. **Find the circular step.** A student is asked to prove the Vertical Angles Theorem and writes the proof below.

Given: $\angle AQC$ and $\angle BQD$ are vertical angles. **Prove:** $\angle AQC \cong \angle BQD$.

Statements	Reasons
$\angle AQC$ and $\angle BQD$ are vertical angles	Given
$m\angle AQC = m\angle BQD$	Vertical angles are congruent
$\angle AQC \cong \angle BQD$	Definition of congruent angles

Which step is circular, and why? Then name the two facts about linear pairs that a correct proof would use instead.

Answer: ________________

2. Which reason works? Point M lies between A and C on $\overline{AC}$. A proof needs the statement $AM + MC = AC$. Three reasons are offered:

(i) Segment Addition Postulate

(ii) Because $AM + MC = AC$

(iii) Because M is the midpoint of $\overline{AC}$

Choose the reason that legitimately supports the statement, and say what is wrong with each of the other two — one of them is circular, the other assumes something the givens never state.

Answer: ________________

3. Find the mistake. Points A, Q, and C are collinear and ray QB is drawn, so $\angle AQB$ and $\angle BQC$ form a linear pair. It is given that $m\angle AQB = 3x + 10$ and $m\angle BQC = 5x - 30$. Dana writes "$m\angle AQB = m\angle BQC$, because corresponding angles are congruent", solves $3x + 10 = 5x - 30$, and reports $x = 20$ and $m\angle AQB = 70$.

(a) What does Dana's reason assume that the givens never state?

(b) Use the linear pair to find x, then find $m\angle AQB$.

Answer: ________________ degrees

4. Find the mistake. Point M lies between A and B. It is given that $AM = 2x + 3$, $MB = 4x - 7$, and $AB = 20$.

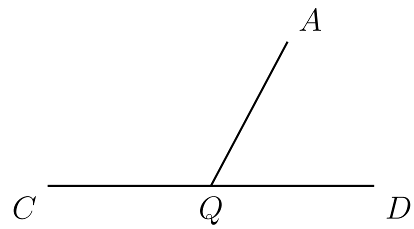
Ben writes "$AM = MB$, because M is the midpoint of $\overline{AB}$", solves $2x + 3 = 4x - 7$, and reports $x = 5$ and $AM = 13$.

(a) Which given would Ben need before he could write $AM = MB$? Is it there?

(b) Use the Segment Addition Postulate to find x, then find AM.

Answer: ____________________

5. Find the mistake. In the diagram, C, Q, and D are collinear and ray QA is drawn. It is given that $m\angle AQC = 5x - 12$ and $m\angle AQD = 3x + 8$.



Priya writes "$\angle AQC \cong \angle AQD$, because vertical angles are congruent", and reports $x = 10$ and $m\angle AQC = 38$.

(a) Explain why these two angles cannot be vertical angles, and name the relationship they actually have.

(b) Use that relationship to find x, then find $m\angle AQC$.

Answer: ________________ degrees

6. Repair the argument. In $\triangle ABC$ it is given that $\angle A \cong \angle B$. A student argues: *"$\overline{CA} \cong \overline{CB}$ because the triangle is isosceles, and the triangle is isosceles because its base angles $\angle A$ and $\angle B$ are congruent."*

(a) Explain exactly why this argument is circular.

(b) Write a correct two-column argument (statements and reasons) that reaches $\overline{CA} \cong \overline{CB}$ from the given.

Answer: ______________